Laboratory of Advanced Programming

Project Presentation ~ A.Y. 2025/2026

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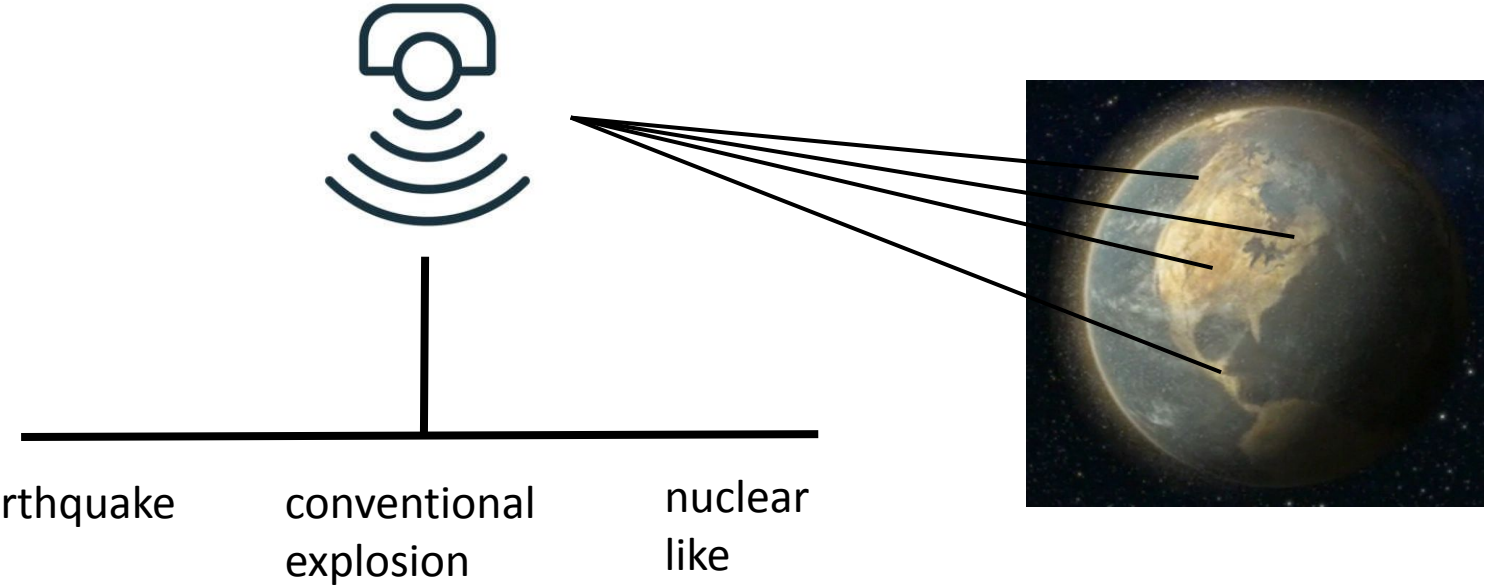
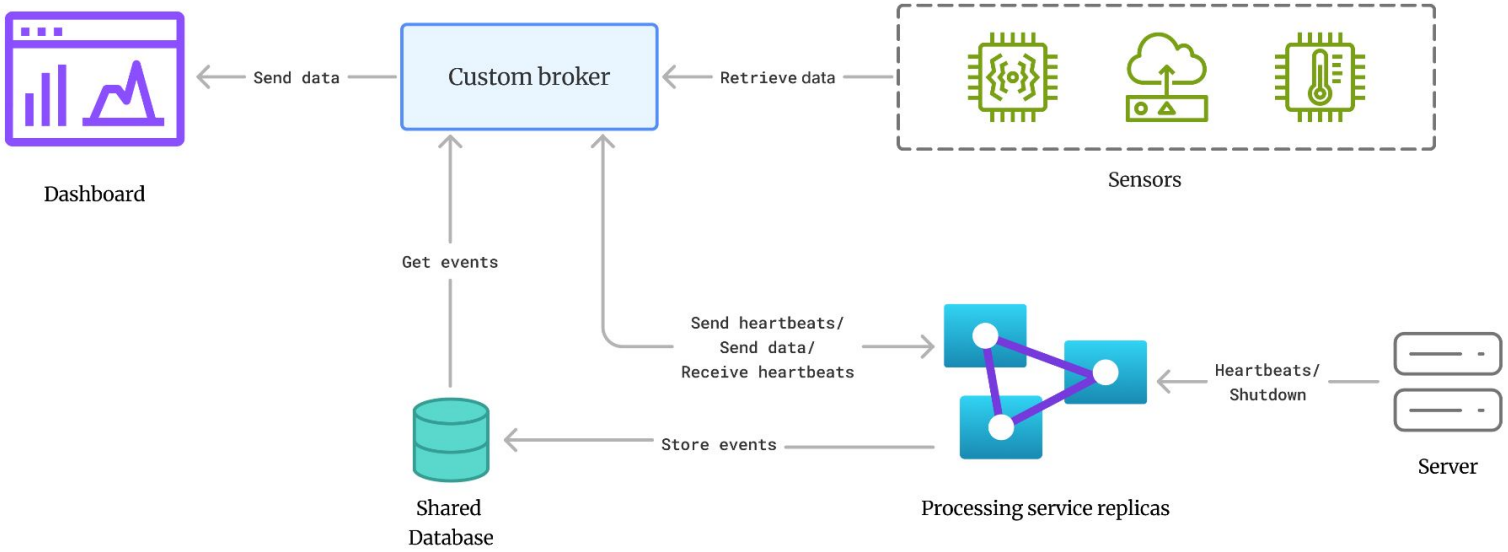
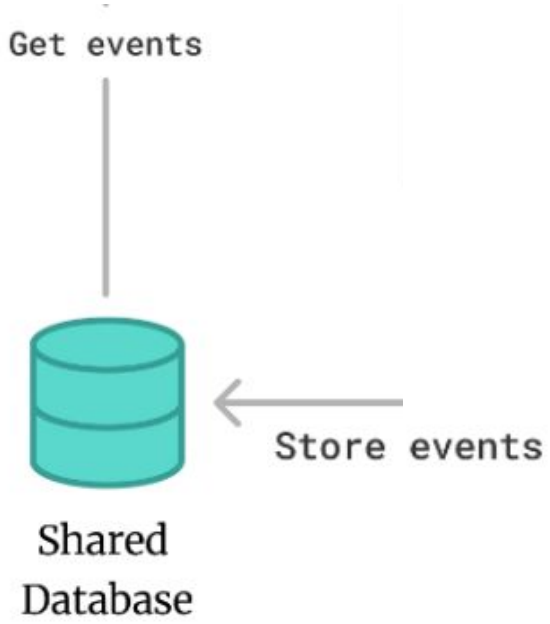
A Project to save the F***ing World!



Architecture overview



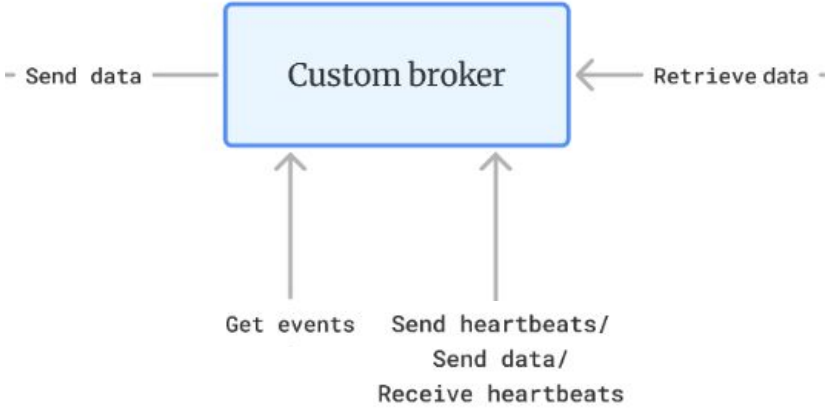
The Database



- Postgres:
  - structure of the database
- Events storage:

```sql
CREATE TABLE IF NOT EXISTS events (
    sensor_id TEXT NOT NULL,
    event_type TEXT NOT NULL,
    startstamp TIMESTAMPTZ NOT NULL,
    endstamp TIMESTAMPTZ NOT NULL,
    frequency DOUBLE PRECISION NOT NULL,
    PRIMARY KEY (sensor_id, startstamp, endstamp)
);
```

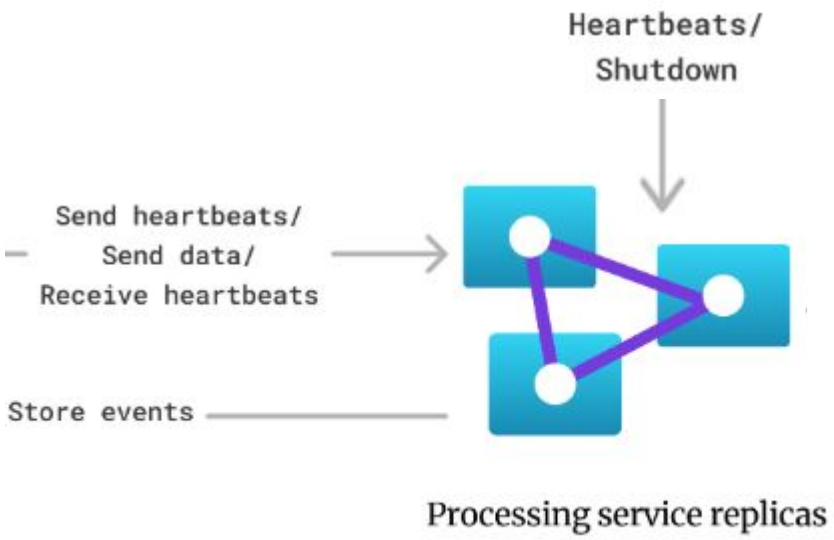
The Custom broker



We implemented a master-slave architecture; the *custom broker* is the master.

- multi-threading
- reads data from the simulation server, and forwards them to the replicas
- checks when replicas fail
- sends sensor's measures to the dashboard
- communication with simulation server and dashboard:
  - websockets
- communication with replicas:
  - regular sockets
- elects a replica as leader

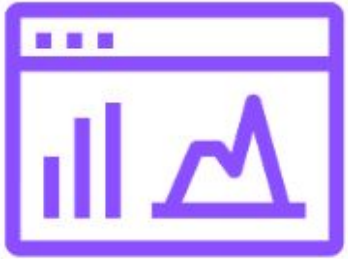
The Processing service Replicas



Processing service replicas (slaves)

- replicated
- process data received from the master and thus detect events
- may fail
- the elected leader:
  - performs the frequency analysis
  - stores classified events in the database
- receives commands from the simulation server

The Dashboard

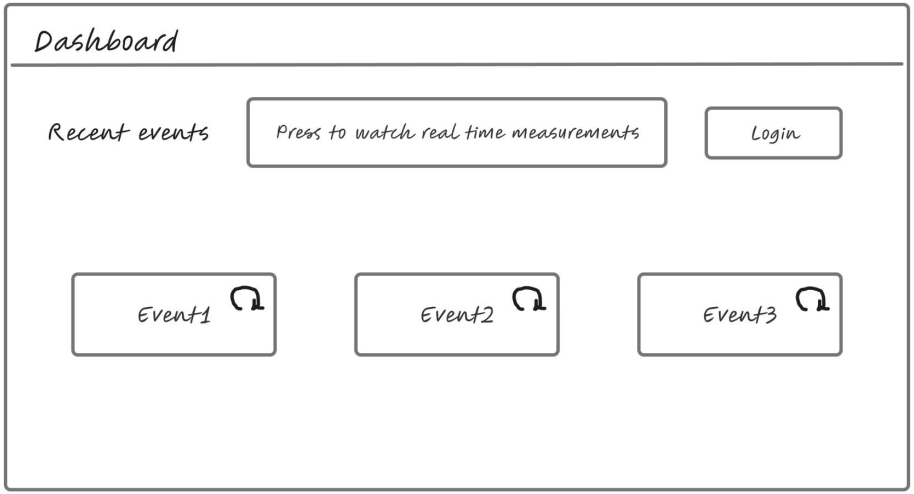


Send data

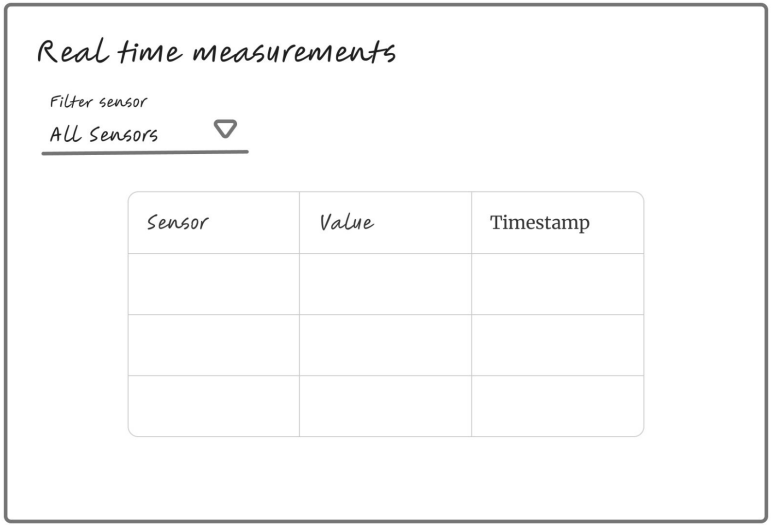
Dashboard

- NiceGUI library :
  - structure of the interface
- Websocket :
  - communication with broker for real time data
- Psycopg2 :
  - communication with database for history data

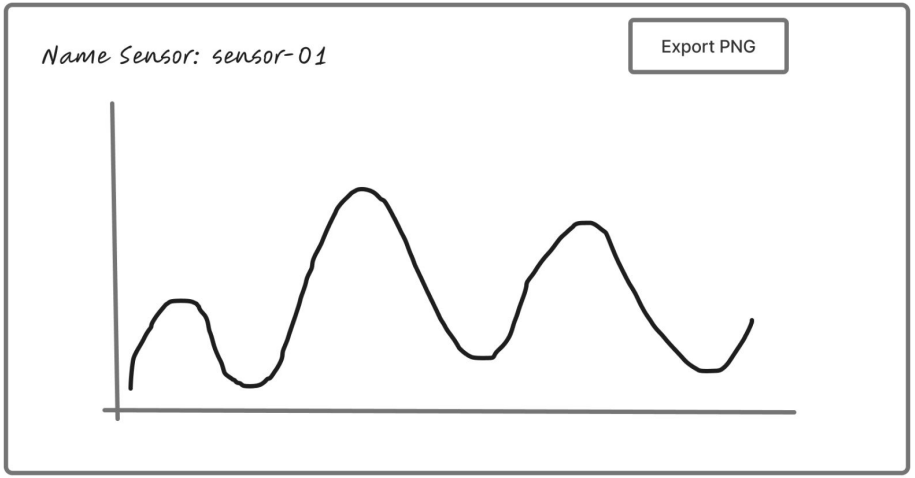
Mockups


Dashboard
Recent events
Press to watch real time measurements
Login
Event1
Event2
Event3

Mockups

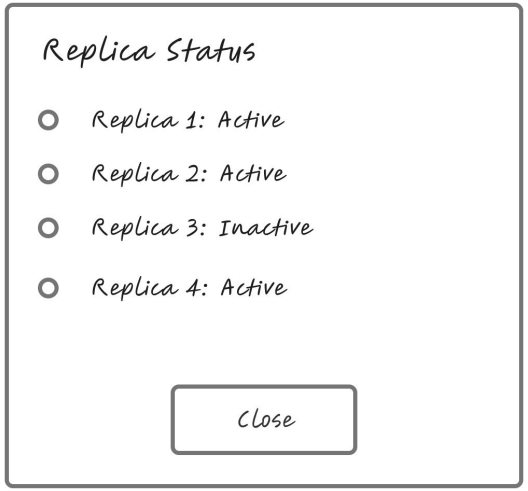

Real time measurements
Filter sensor
All Sensors
Sensor
Value
Timestamp

Mockups



Mockups

Mockups


Replica Status
Replica 1: Active
Replica 2: Active
Replica 3: Inactive
Replica 4: Active
Close

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